

Exp No: 5 - ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

AIM:

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus

SOFTWARE REQUIRED:

- Proteus 8 software
- Arduino IDE

PROGRAM:

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;  
int delay_time=200;
```

```
void setup() {  
  //configure pin2 as an input and enable the internal pull-up resistor  
  pinMode(12, INPUT_PULLUP);  
  pinMode(1,OUTPUT);  
  pinMode(2,OUTPUT);  
  pinMode(3,OUTPUT);  
  pinMode(4,OUTPUT);  
  pinMode(5,OUTPUT);  
  pinMode(6,OUTPUT);  
  pinMode(7,OUTPUT);  
}
```

```
}  
void loop() {  
  //read the pushbutton value into a variable  
  int sensorVal = digitalRead(12);
```

```
  if (sensorVal == LOW) {  
    x0=true;  
  }  
  while(x0){  
    zero();  
    sensorVal = digitalRead(12); if  
    (sensorVal == LOW) {  
      x1=true;  
      x0=false;  
    }  
  }  
  while(x1){  
    one();  
    sensorVal = digitalRead(12); if  
    (sensorVal == LOW) {
```

```
x2=true;
x1=false;
}
}
while(x2){
two();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x3=true;
x2=false;
}
}
while(x3){
three();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x4=true;
x3=false;

}
}
while(x4){
four();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x5=true;
x4=false;
}
}
while(x5){
five();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x6=true;
x5=false;
}
}
while(x6){
six();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x7=true;
x6=false;
}
}
while(x7){
seven();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x8=true;
```

```

x7=false;
}
}
while(x8){

eight();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x9=true;
x8=false;
}
}
while(x9){
nine();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x0=true;
x9=false;
}
}

}

void zero()
{
for(int i=1;i<7;i++)
{
digitalWrite(i,HIGH);
digitalWrite(7,LOW);
}
delay(delay_time);
}

void one()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);

digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,LOW);
delay(delay_time);
}

```

```
void two()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,LOW);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time); }
```

```
void three()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time); }
```

```
void four()
{

digitalWrite(1,LOW);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
```

```
void five()
{
digitalWrite(1,HIGH);
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
```

```
delay(delay_time); }
```

```
void six()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);
```

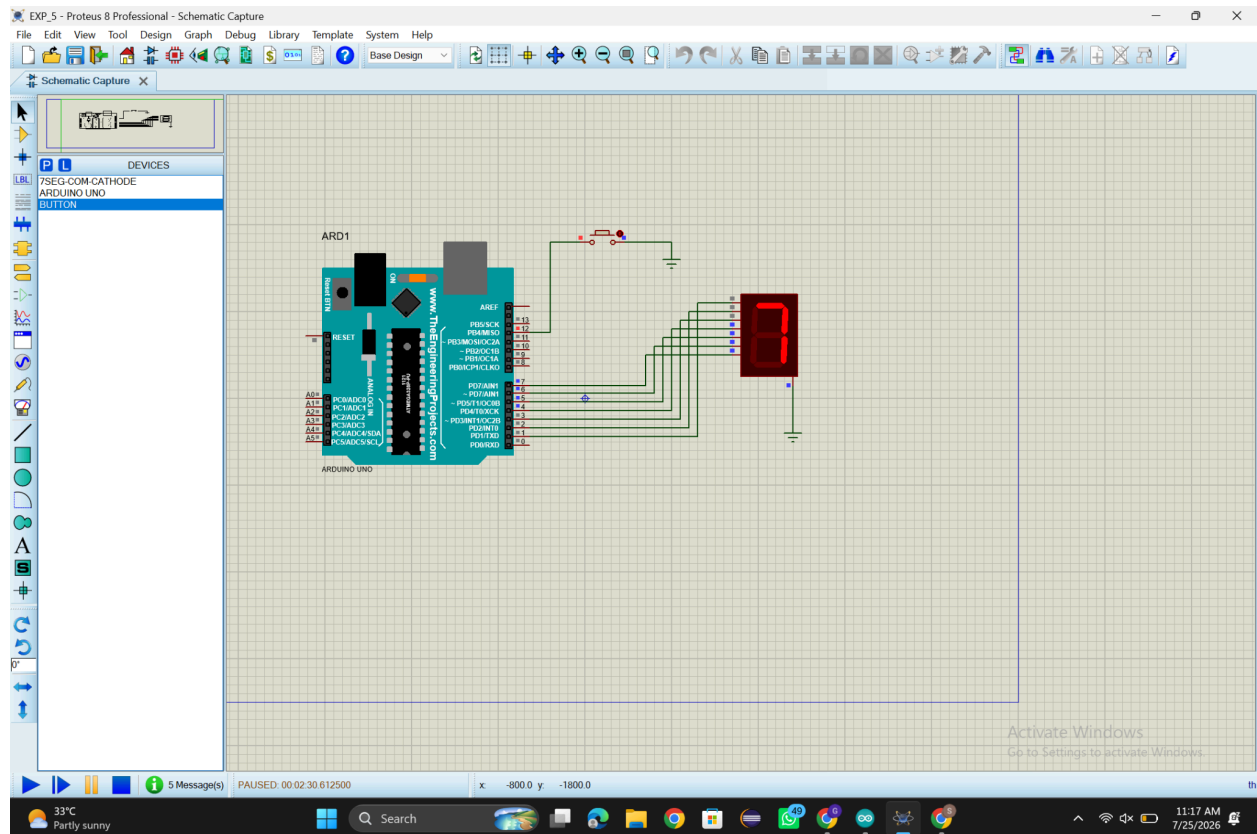
```
}  
void seven()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,LOW);  
delay(delay_time); }
```

```
void eight()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void nine()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);
```

```
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.